Leadership Plan

The MOSAIC leadership team has extensive experience directing large, multi-investigator research programs, interdisciplinary modeling efforts, and collaborative NIH-funded initiatives. Papin (PI of MOSAIC) has served as PI of an NIH T32 training program (with 35+ faculty mentors and 60+ trainees) for nearly ten years. He served as Director and PI of an NIH KL-2-funded faculty development program (with 40+ faculty across three institutions) for 5 years. In addition to other NIH, NSF, DOD grants, he is currently a co-lead of a project in an NIH U54-funded program. With his leadership as Editor-in-Chief of the journal PLOS Computational Biology and as an elected member of the Board of Directors of the Biomedical Engineering Society, he has leadership experience with large complex scientific organizations (100s of members of journal editorial board; 1000s of society members). Peirce-Cottler has been a member of a multi-institutional consortium funded by the Allen Foundation (with Agmon) on multiscale models of bacteria. Derré is currently working with an NIH U54-funded program for an “organ-on-a-chip” program for the female reproductive tract. Khetani is a part of a NIH UM1-funded consortium on New Approach Methodologies (NAMs) on chemical safety assessment. Agmon has worked with several NIH and DARPA funded multi-institutional programs on biomedical modeling. Young, Mac Gabhann, Arnold, and Kolling also have extensive experience with NIH U, P, and related grant programs.

There is a rich history of collaboration among team members. Papin and Mac Gabhann served as co-Editors-in-Chief of PLOS Computational Biology. Papin, Mac Gabhann, Peirce-Cottler, Arnold, and Agmon have an extensive collaboration history with multiple co-authored manuscripts and co-led grant applications among them. Papin and Derré have collaborated through academic advising roles. Khetani/Papin and Kolling/Young have long histories of scientific collaboration on grant and project development.

The Investigators will be organized as outlined in Figure 1. All Investigators, trainees, and staff will be a part of one of four scientific teams. Arnold will lead the BV team; Peirce-Cottler will lead the MASLD team; Mac Gabhann will lead the Integration team; and Agmon will lead the Software team. Each modeling-focused lab will have a lab member on each of the different teams. All experimental/clinical team members will be associated with the BV or MASLD groups according to their respective expertise. Team leads will report progress, challenges, and milestone attainment to Papin and Kolling during the monthly project meetings. These meetings will be used to coordinate activities across the BV, MASLD, Integration, and Software teams and ensure timely achievement of project milestones.

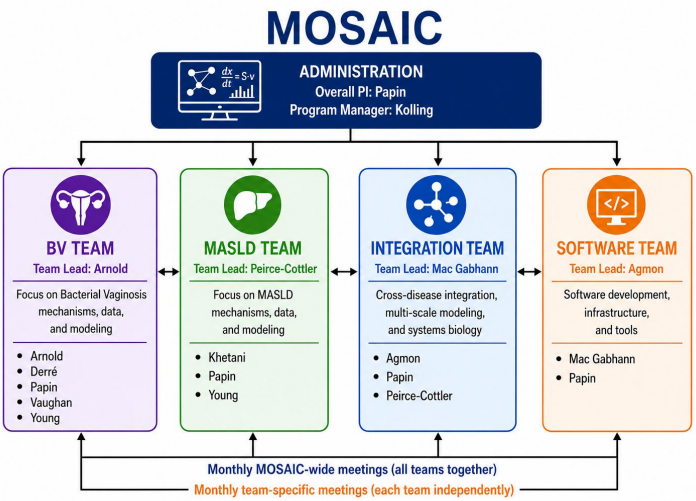


Figure 1. MOSAIC Leadership Structure

All project members will attend monthly online meetings to communicate project objectives, results and challenges. Project teams will meet separately on a monthly basis to discuss team-specific progress. All project members will communicate weekly as needed, either by Slack, phone, e-mail, or in-person to discuss smaller project-related matters. An annual in-person meeting will be held at the University of Virginia to present milestone accomplishments and discuss future objectives of the project among all team members. Kolling will serve as the program manager (with 50% effort committed) and will coordinate all communications.

The University of Virginia will establish sub-contracts with the respective institutions where each Investigator will be responsible for their own fiscal and research administration. Papin will provide overall scientific leadership for MOSAIC and will report all progress and necessary documents to the NIH, including progress reports and IRB approvals. Any scientific or administrative conflict will be addressed by the project leadership team and, if necessary, elevated to institutional leadership for resolution. If an Investigator moves to a new institution, attempts will be made to transfer the relevant portion of the grant to the new institution. In the event that an Investigator cannot carry out their duties, a new Investigator will be recruited as a replacement at one of the participating institutions.

Abstract

Sex-specific differences in therapeutic efficacy and toxicity are pervasive, yet the mechanistic role of dynamic hormone homeostasis in shaping these differences remains poorly understood. Current computational models fail to capture how variations in hormone regulation, metabolism, and host-microbiome interactions propagate across biological scales to alter drug performance. To bridge this critical gap, we will develop **MOSAIC** (**M**odeling **O**f **S**ex-specific metabolism **A**cross **I**nteracting **C**ommunities), an open-source, multi-scale computational framework designed to capture and predict hormonal, metabolic, and microbiome states across interacting organ systems. **MOSAIC** addresses the challenge of cross-scale physiological modeling by defining mechanistic interfaces that allow diverse, independently developed modeling paradigms to interact dynamically without sacrificing specific biological assumptions. A central focus is modeling microbiome-mediated estrogen metabolism and enterohepatic recycling as a systemic network linking gut, liver, and vaginal physiology. To ensure scalability and extensible curation, we will deploy innovative, AI-assisted tools to automate data harmonization, integrate existing multi-omic datasets, parameterize sex-specific metabolic networks, and prioritize simulation results for experimental validation. The key achievable goals will be tol: 1) Develop a multi-scale model of the hormone-regulated vaginal microbiome to predict local therapeutic interventions, 2) Build an integrated, sex-specific computational model of gut–liver host–microbiome metabolism incorporating hormone homeostasis to predict systemic therapeutic efficacy, and 3) Integrate the gut–liver and vaginal models to simulate system-level feedback, hormone-mediated transport, and drug-processing across compartments. Using bacterial vaginosis (BV) and metabolic dysfunction-associated steatotic liver disease (MASLD) as demonstration systems, **MOSAIC** will deliver the first human-relevant *in silico* platform capable of predicting therapeutic efficacy and adverse effects in hormone-regulated systems, ultimately accelerating precision medicine strategies.

Specific Aims

Evaluating how drugs and biologics perform across diverse patient populations is severely limited by our current inability to computationally map the systemic influence of sex hormones. While sex-specific variations in therapeutic efficacy and toxicity are widely acknowledged, the dynamic, system-level mechanisms driving these differences remain elusive. Existing metabolic models are restricted to single human cell types or localized microbial niches, failing to capture the physiological crosstalk linking host endocrine regulation, metabolic fluxes, and microbial ecology. Because perturbations in one compartment inevitably cascade across organ systems, **there is a critical need for a human-relevant, mechanistic framework** that treats hormone homeostasis as a dynamic regulator of host-microbiome and metabolic networks.

To bridge this critical gap, our overall objective is to develop, validate, and disseminate **MOSAIC** (**M**odeling **O**f **S**ex-specific metabolism **A**cross **I**nteracting **C**ommunities)**. MOSAIC** is an open-source, multi-scale computational framework that integrates host tissue metabolism, microbiome composition, and dynamic sex-hormone homeostasis to predict therapeutic efficacy and toxicity. Our **central hypothesis** is that explicit mechanistic interfaces between disparate modeling paradigms will allow independently developed models to function as an integrated representation of hormone-regulated human physiology. Our team is uniquely positioned to pioneer this approach using innovative modeling approaches with AI-based tools to automate data harmonization, integrate existing clinical multi-omic datasets, quantify parameter uncertainty, and prioritize high-value simulation results for experimental validation. We will demonstrate and validate this framework using publicly available datasets and *in vitro* approaches as needed to examine bacterial vaginosis (BV) and metabolic dysfunction-associated steatotic liver disease (MASLD) – two highly prevalent conditions with profound sex-specific components driven by host-microbiome-hormone-metabolism cross-talk – in the context of estrogen dynamics. The initial implementation of **MOSAIC** is focused on estrogen-mediated physiology in BV and MASLD as tractable demonstration systems. However, the framework is designed for extension to hormone-modulated diseases. Objectives will be achieved with the following aims:

Aim 1: Develop a multi-scale computational model of the hormone-regulated vaginal microbiome to predict local therapeutic efficacy. We will integrate ecological and metabolic models of vaginal microbial communities to assess how systemic metabolic and hormonal states shape local mucosal environments and community stability. Utilizing AI-assisted tools for model curation and parameterization, we will test the model's ability to predict the efficacy and adverse-effect risk of standard antibiotics and emerging biologics for BV across varying patient-specific hormonal states.

Aim 2: Construct an integrated, sex-specific model of gut–liver–microbiome metabolism under hormonal regulatory control. We will build a multi-compartment modeling framework that couples genome-scale representations of male and female hepatic metabolism with community metabolic models of the gut microbiome. The framework will be parameterized using human transcriptomic, metabolomic, and metagenomic datasets to capture hormone-regulated metabolic fluxes. We will validate the model using *in silico* and *in vitro* approaches of drug-processing pathways and therapies in the context of MASLD.

Aim 3: Integrate the gut–liver and vaginal models to predict system-wide, compartment-coupled therapeutic modulation. We will connect the models from Aims 1 and 2 to explicitly simulate the synthesis, microbial deconjugation, enterohepatic recycling, and systemic transport of sex hormones across biological compartments. We will predict how systemic variations in microbiome composition and endocrine status propagate across organ systems to alter downstream metabolic network behavior and patient-specific therapeutic responses.

Expected Outcomes and Impact: Upon completion, this work will deliver the first human-focused *in silico* framework capable of explicitly modeling hormone homeostasis as a dynamic, system-level regulator of host-microbiome interactions. By providing an open, modular, and reusable platform, MOSAIC will transform how the scientific community investigates sex-specific physiology, driving the accelerated design of precision therapeutic strategies.

Public Health Relevance

Tens of millions of women per year in the United States are affected by bacterial vaginosis and metabolic dysfunction-associated steatotic liver disease (MASLD), two conditions strongly influenced by dynamic hormone regulation and interconnected metabolic and microbiome processes. Progress has been hindered by limited understanding of the complex interplay among hormonal, physiological, metabolic, and microbiome systems. This program will develop mechanistic computational models to uncover the biological mechanisms underlying these interactions and to predict sex-specific therapeutic efficacy and toxicity, ultimately enabling improved therapeutic strategies for women’s health.

MOSAIC: A mechanistic, multi-scale modeling framework for hormone-regulated host-microbiome metabolism to predict sex-specific therapeutic efficacy and toxicity

MOSAIC (**M**odeling **O**f **S**ex-specific metabolism **A**cross **I**nteracting **C**ommunities) addresses a fundamental barrier in computational modeling for clinical applications: sex-specific differences in therapeutic response often arise from interactions across biological systems that are still modeled in isolation. Hormones, metabolism, microbial communities, tissue physiology, immune state, and drug exposure form a coupled physiological network, but current models rarely capture how changes in one subsystem propagate through the others to alter efficacy or toxicity. We will develop an open, modular modeling framework that connects these mechanisms across scales while preserving the assumptions, data sources, and biological scope of each component model. By making these interactions executable, shareable, and reusable, **MOSAIC** will enable mechanistic predictions of how hormone-regulated host-microbiome dynamics shape therapeutic response in bacterial vaginosis (BV), metabolic dysfunction-associated steatotic liver disease (MASLD), and future disease contexts where sex-specific physiology remains an unresolved driver of clinical variability.

Individuals with similar physiological measurements can experience markedly different clinical outcomes. We focus on two conditions for which the complex interplay between hormone homeostasis, metabolism, and the microbiome manifests in sex-specific ways and for which responses to therapeutic interventions are difficult to predict. First, BV affects more than 20 million women in the US (1). It arises from dysbiosis of the vaginal microbiome where the interaction between hormone dynamics and host environment results in enormous challenges to resolve the condition (2). Second, MASLD affects nearly 100 million people in the US and nearly half of affected individuals are women, with marked differences between pre- and post-menopausal populations (3,4). MASLD arises from a complex interplay between liver metabolism, gut microbial ecology, immune responses, and hormone regulation, with perturbations in one subsystem propagating across scales to influence disease progression (5). Treatment strategies for these two conditions can vary widely in efficacy. For example, two women receiving the same treatment for BV may experience different recurrence patterns due to interactions among vaginal microbial ecology, estrogen-regulated epithelial physiology, immune responses, metabolic profiles, and antibiotic exposure (6). Existing computational models are typically designed to address individual components of these systems—for example, predicting drug exposure, microbial community dynamics, or metabolic activity—but they rarely capture how these processes interact across biological scales. As a result, important outcomes, including therapeutic efficacy and toxicity, may emerge not from any single subsystem but from feedback among hormones, metabolism, microbial communities, immune responses, and therapeutic interventions. **MOSAIC** is designed to make these interactions executable, enabling mechanistic explanations and predictions of heterogeneous responses to therapeutics in hormone-regulated physiology. The breakthrough is not any single component model, but the executable, hormone-centered coupling layer that allows independently validated metabolic, microbiome, PKPD, and agent-based models to propagate sex-specific therapeutic effects across tissues.

Scientific rationale and justification of the approach

Sex-specific differences in therapeutic efficacy and toxicity are pervasive, yet the mechanistic role of dynamic hormone homeostasis in shaping these differences remains poorly understood. Current computational models do not capture variability linked to hormone regulation of metabolism and host-microbiome interactions, limiting predictions on how drugs and biologics perform across sex-specific physiological states. To address this gap, we will develop a computational framework to capture hormonal, metabolic, and microbiome states in multi-scale computational models that predict therapeutic efficacy and toxicity. This framework uniquely integrates genome-scale metabolic models of human and microbial physiology (Papin), dynamic microbiome models (Arnold), multicellular agent-based models (Peirce-Cottler), whole-body compartment models (Mac Gabhann), and robust software design of integrative modeling frameworks (Agmon), enabling mechanistic assessment of the role of metabolism, microbial species, and sex hormones in therapeutic efficacy. These models may also resolve open questions about the causal drivers vs. correlative factors associated with sex-specific differences in disease. We will implement this framework as an open-source platform termed **MOSAIC**. We will leverage team expertise in all these modeling methods as well as domain-specific expertise in hepatic (Khetani) and vaginal (Derré) cell biology, along with clinical expertise in the gastrointestinal (Young) and vaginal (Vaughan) microbiome.

As an initial proof-of-concept, we will focus on BV, a condition in which hormone-regulated shifts in the vaginal microbiome and metabolic environment directly influence therapeutic response to antibiotics and

emerging biologics (probiotics). We will extend the framework to MASLD, where gut microbiome-derived metabolites and hormone-regulated hepatic metabolism influence sex-specific therapeutic efficacy and toxicity, with reported sex-specific differences in efficacy of existing drugs including GLP-1 receptor agonists as examples (7,8). Despite both BV and MASLD being at the interface of metabolism and the microbiome, these prevalent conditions with clear sex-specific differences are typically studied in isolation.

A key driver linking metabolism and the microbiome, particularly in the context of these diseases, is systemic hormone homeostasis (9,10). For example, the gut microbiome contributes to systemic estrogen availability through microbial deconjugation and enterohepatic recycling, while estrogen modulates hepatic metabolism, immune responses, and vaginal epithelial physiology (11–14). Perturbations in the dynamics of estrogen and other sex-specific hormones—arising from microbiome composition, body composition, dietary inputs, or reproductive stage—have the potential to propagate across organ systems. These interactions suggest that hormone-regulated metabolism represents a unifying lens to examine seemingly disparate disease states and dysbiotic conditions. Although the initial implementation focuses on estrogen biology because of its central role in BV and MASLD, the framework is designed to accommodate additional endocrine signals including androgen, thyroid, glucocorticoid, and peptide hormone signaling. BV and MASLD are used as demonstration systems because both exhibit strong dependence on hormone-regulated physiology. The primary objective of **MOSAIC** is not disease-specific prediction, but rather the development of a reusable framework for understanding how perturbations in hormone homeostasis propagate across biological scales to influence therapeutic response.

MOSAIC will enable identification of potential therapies for metabolic and microbial targets and prediction of adverse effects in a sex-specific manner. It will also allow for the integrated modeling of the gut, liver, and vaginal systems and explicitly represent synthesis of sex-specific hormones, microbial metabolism, transport, and enterohepatic recycling, enabling simulation of hormone-mediated feedback across physiological compartments. This model will output predictions about the roles of liver and microbiome metabolism in processing other drugs. Sex-specific physiology will be incorporated through differential parameterization of metabolic networks and hormone dynamics, enabling patient-specific models (digital twins).

Most existing metabolic and dynamic models focus on individual cell- or tissue-types or isolated microbial species. Rarely do models incorporate dynamic hormone regulation or sex-specific physiology. As a result, there is no human-relevant, mechanistic framework for exploring how microbiome-mediated changes in hormone homeostasis influence metabolic fluxes, microbial ecosystem stability, or sex-specific responses to therapeutic interventions. **Our approach represents the first attempt to explicitly model hormone homeostasis as a dynamic, system-level regulator of host-microbiome metabolic interactions across multiple organ systems (Figure 1).** These models will require innovation in AI-based tools for model curation and parameterization, multi-scale model coupling, and data integration to capture sex-specific human physiology.

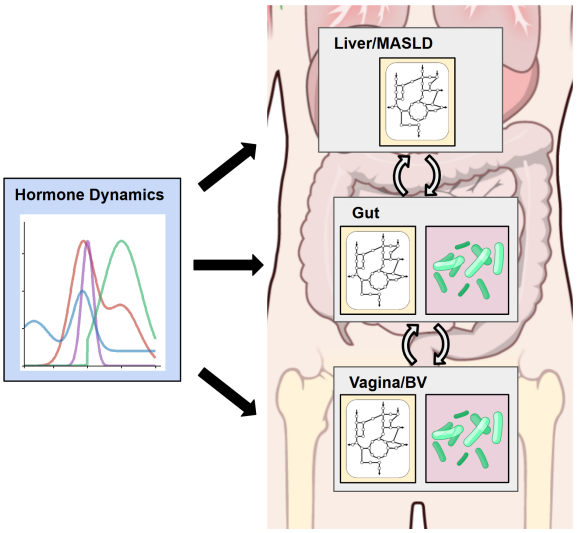


Figure 1. MOSAIC makes predictions across three tissue types: the liver, the gut, and the vagina. Each tissue type will be modeled by a genome-scale metabolic model, and the gut and vaginal tissues will include additional relevant genome-scale metabolic models, mechanistic community models, and agent-based models according to their specific microbial community. Together, these compartments will be used to study how MASLD in the liver and BV in the vagina affect metabolism and microbiome composition in these tissues.

To enable scalable and extensible modeling, **MOSAIC** will incorporate AI-based tools to integrate and parameterize existing multi-omic datasets (some of which were recently developed in our own labs), automate data harmonization, quantify uncertainty, and facilitate model curation. Our labs are pioneering the application of AI-based tools to mechanistic computational models. AI-based tools will also enable the identification of the most informative simulation results to prioritize experimental validation efforts. This work will deliver an innovative, human-focused *in silico* framework capable of supporting hypothesis generation, model refinement, and broad community use as a reusable platform to study hormone-regulated host-microbiome interactions.

Short-term (first year) objectives

- Curate existing data for MASLD and BV model construction
- Implement AI-based data curation, integration, and parameterization tools
- Construct a hormone-regulated vaginal microbiome digital twin model
- Establish model credibility criteria and validation pipeline

Long-term (second and third year) objectives

- Construct a sex-specific, hormone-regulated gut-liver-microbiome digital twin model
- Integrate vaginal and gut-liver modules into a unified **MOSAIC** platform
- Enable patient-specific, sex-specific therapeutic predictions
- Identify mechanistic therapeutic targets and prioritize experimental validation
- Foster community adoption and extensibility

Description of the proposed model

MOSAIC provides a mechanistic, quantitative approach to understand how individual microbial species and community-wide metabolic networks collectively respond to sex-specific hormone levels and dictate therapeutic responses. A core innovation of our framework lies in its automated, individualized pipeline. Each digital twin produced by **MOSAIC** will provide an accurate reflection of the real-time ecological forces shaping an individual's microbiome environment under varying lifestyle factors and sex-specific transcriptomics. Ultimately, we will leverage this digital twin framework to 1) predict patient-specific therapeutic efficacy and adverse drug reactions before treatment administration, 2) design precise nutritional, biotherapeutic, or hormonal interventions that restore homeostatic balance and optimize drug response, and 3) elucidate sex-specific differences in the dynamic bidirectional networks driving these responses. By providing a rigorous, automated, and scalable mathematical foundation for digital twins, this work shifts the paradigm from descriptive association studies to predictive, personalized medicine—paving the way for targeted therapies that account for sex-specific differences and optimize both endocrine health and drug efficacy. The integrative framework for **MOSAIC** is outlined below:

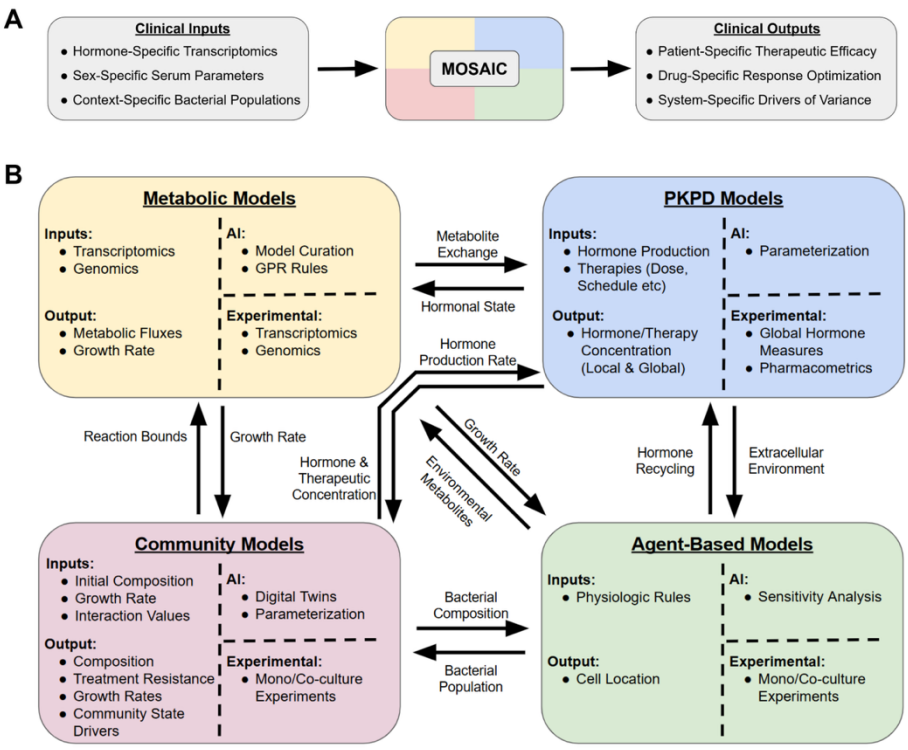


Figure 2. (A) MOSAIC can use several relevant clinical inputs, such as hormone-specific transcriptomics, sex-specific serum-derived parameters, and context-specific bacterial populations, and produce many important clinical outputs, such as hormone-specific therapeutic effects, sex-specific metabolic changes, and context-specific microbiome population shifts. (B) Each modeling approach has relevant inputs, outputs, AI usage, and in vitro experiments. Importantly, outputs from one approach can serve as a way to improve assumptions and inputs to other modeling approaches, meaning changes to one approach propagate out to influence other modeling approaches.)

*Context-specific genome-scale **metabolic models***

We will create genome-scale metabolic models (GEMs) for each human tissue and microbiome community in a high-estrogen and low-estrogen environment. The liver and gut models will be created from Human1 , a previously published general human cell GEM, and publicly available transcriptomic data (see data resources section below) in combination with fast task-driven integrative network inference for tissues (ftINIT) (15), which will inform which reactions are most important for the given tissue type and hormone environment. We will use transcriptomic data for the microbial

species in environments that simulate the associated physiological niches (gut, vagina), with particular attention to *Gardnerella* and *Lactobacillus* species for the vaginal microbiome (16) and *Klebsiella*, *Escherichia*, *Streptococcus*, *Faecalibacterium*, and *Roseburia* species for the gut microbiome (17) given their known associations with BV and MASLD, respectively. As needed, we can also generate associated transcriptomics data with *in vitro* conditions that simulate these physiological environments. Metabolic network reconstructions are readily generated at these scales with the software tool Reconstructor (18) which we developed. LLMs will be used to curate gene-protein-reaction rules in the GEMs to improve the prediction of gene perturbations and drug targets with a pipeline we recently developed. We will sample the feasible flux distributions in these tissue- and environment-specific models to predict metabolic reaction activity and growth rates. AI-based random forest or LASSO tools will be used to select a sparse set of discriminatory reactions from high-dimensional flux datasets and identify those most predictive of different physiological states. Papin's team has used these methods previously for human models to successfully predict inflammatory metabolites in Crohn's disease (19) and targetable genes in gastrointestinal bacteria (20).

Mechanistic microbiome ***community models***

Existing computational approaches to study the microbiome are siloed. Ordinary Differential Equation (ODE) models excel at capturing the macro-level, population-level dynamics of microbial communities over time, but manual parameterization for individual patients is bottlenecked by data availability and species selection, and internal cellular metabolism is treated as a "black box". Conversely, Genome-Scale Metabolic Models (GEMs) provide high-resolution, constraint-based reconstructions of intracellular metabolic pathways, but they struggle to inherently capture time-varying population shifts and complex intercellular dependencies in a dynamic ecosystem. This proposal addresses these critical methodological gaps by introducing an innovative, multiscale computational framework designed to construct microbiome digital twins. Arnold's team has pioneered the development of microbiome community models for the prediction of microbial dynamics in BV (21–23). This proposed approach utilizes data-driven models to capture personalized species composition and parameterize dynamic ODEs, subsequently isolating and prioritizing key microbial interactions for high-resolution metabolic evaluation using Genome-Scale Metabolic Models (GEMs). This framework bridges the gap between macro-scale ecosystem dynamics and micro-scale metabolic fluxes to relate host hormone bioavailability to drug efficacy with unprecedented precision.

Pharmacokinetic/pharmacodynamic ***(PKPD) models***

We will construct a multi-compartment whole-body pharmacokinetic (PK) model that includes compartments representing the key organs for BV and MASLD (e.g., gut, liver, kidney, female reproductive tract), connected via a central blood compartment, to track systemic and local levels of key sex hormones and therapeutics. The PK model will mechanistically represent estrogen biosynthesis, therapeutic delivery dosage (including route, dose, and time), tissue distribution, and hepatic/renal clearance, parameterized with sex-specific physiological data. Each molecular species is tracked by one equation in each compartment, and the terms in these equations are the quantitative rates representing each biological mechanism that affects that molecule. Mac Gabhann has pioneered such approaches for sex-specific prediction of cell proliferation in the female reproductive tract (24). Drug-based therapies, such as antibiotics (e.g., metronidazole) for BV or resmetirom for MASLD, will be modeled with oral delivery, where both the local and systemic effects following absorption will be modeled. Non-drug therapies, such as biologics (probiotics like *L. crispatus*) for BV, will be modeled with delivery of the probiotics into the community model, from which the resulting modified production of hormones will be an input to the PK model. The pharmacodynamic (PD) component will couple predicted plasma estrogen concentrations to estrogen receptor α/β and therapy-specific binding kinetics in target tissues, plus the impacts of exogenous drugs on the hormone system. The PKPD model takes as input the hormone production levels from the community and agent-based models and therapeutic dosing based on the specific clinical interventions being modeled. The outputs of the PKPD model include systemic and local hormone and drug levels, which serve as inputs to the community model, agent-based model, and metabolic model, quantifying the hormone and drug environment in which each of those models are set. Outputs from this model can be used to inform what kind of hormonal environment would be most appropriate to use in our other models. Experimental measurements of the plasma levels of hormones and drugs are measured as possible so that the predictions of the model can be compared to the clinical pharmacometrics. This direct clinical connection facilitates the parameterization of the models using sex-specific data. Mac Gabhann's team has

created multiple similar PKPD models that simulate not only exogenous therapeutics including biologics such as antibodies (25,26), gene therapies (27), and cell therapies (28,29), but also simultaneously tracking the relevant endogenous molecules that are distributed systemically such as hormones and cytokines (24,30,31), and the PD interactions between the exogenous and endogenous molecules.

Agent-based models

We will implement an agent-based model (ABM) in NetLogo (32) to simulate the spatiotemporal dynamics of bacterial-epithelial interactions in the gut and vaginal microenvironments. Individual bacterial agents and epithelial cells will be assigned stochastic, rule-based behaviors, including adhesion, proliferation, apoptosis, and cytokine secretion, governed by local environmental conditions. Hormonal concentrations (estrogen and progesterone) will be incorporated as diffusible extracellular fields that modulate agent behavior. Machine learning and AI will be applied to optimize agent rules and identify key cellular interactions driving sex-hormone-dependent dysbiosis or homeostasis. Peirce-Cottler framed the importance of modeling complex biological systems, including immune responses (33,34), in terms of interacting agents and emergent properties, establishing a foundation for multiscale modeling and hybrid computational approaches (35). Methodologically, the development of tools such as Vivarium (a collaboration between Peirce-Cottler and Agmon) has enabled flexible integration of different types of modeling approaches, including agent-based components, facilitating reusable, composable simulations of biological systems across scales (36). These tools and approaches have been deployed in recent research applications, such as whole-cell/agent-based simulations of *E. coli* colonies that quantify single-cell heterogeneity in antibiotic responses (37) and spatial transcriptomics-driven multiscale models of *P. aeruginosa* biofilms (a collaboration between Peirce-Cottler and Papin) that connect metabolic states to microenvironmental stressors (38).

Model Integration

Hormone homeostasis provides the principal coupling mechanism among MOSAIC modules, linking microbial metabolism, tissue physiology, and therapeutic exposure across scales. The goal of **MOSAIC** is to leverage the multitude and diversity of outputs from one modeling method as inputs or constraints for the other modeling methods. These modeling interactions, which are summarized in Figure 2, are as follows:

- Metabolic models provide flux samples and growth rates as output. Growth rates can be used to parameterize individual bacterial species or different host cell types in a given context in both mechanistic community models and agent-based models.
- PKPD models provide therapeutic and hormone concentrations as output. These concentrations can be used to inform the hormone state of the metabolic models and as starting concentrations for the mechanistic community models and agent-based models.
- Mechanistic community models provide community composition, treatment resistance, growth rates, and community state drivers as outputs. Community composition can influence flux bounds in metabolic models, hormone production rate in PKPD models, and agent population number in agent-based models.
- Agent-based models provide cell location and agent behavior as output. These outputs can then parameterize key features of the community models and inform how metabolic models exchange metabolites.

Together, these interactions will help predict the efficacy of several therapeutic strategies. As an example for BV, we can include different biologic (probiotic) or pharmaceutical treatments, such as *Lactobacillus crispatus* or metronidazole, in the **MOSAIC** model. These interventions will modulate metabolite exchange in the metabolic models, growth rates in the community models, hormone production rates in the PKPD models, and cell populations in the agent-based models. Species can be ranked according to their effect toward an anti-BV environment in a given hormonal context. As an example for MASLD, we can predict how a drug such as resmetirom or GLP-1 receptor agonists affects liver metabolism in high- and low-estrogen environments. PKPD modeling would describe how the drug's concentration would change over time, thus determining metabolic pathway changes in the metabolic models. These changes would propagate out to the community and agent-based models via a difference in metabolites exported from the metabolic models. The final prediction would be a decrease in lipid metabolism specific to hormonal context.

AI-empowered modeling tools

AI-driven methods are integral to the **MOSAIC** framework and will support model parameterization, rule curation, and analysis of high-dimensional simulation outputs. AI tools will ultimately improve physiological accuracy and enable identification of key biological drivers underlying a highly complex modeling network. For example, in the personalized microbiome community models, we will use an AI-based adaptive parameterization architecture that can be adjusted based on clinical data availability. When longitudinal clinical data is available, we will use a deep personalization approach by employing a novel AI Autoencoder and Neural Ordinary Differential Equation (Neural ODE) pipeline. A variational autoencoder (VAE) will first compress high-dimensional, noisy clinical time-series data into a low-dimensional latent space representing the underlying ecological states and Lotka-Volterra (LV) model parameters. A Neural ODE network will then learn the continuous-time dynamics within this latent space. By passing these learned gradients back into our mechanistic generalized LV framework, we can efficiently infer high-confidence, individualized interaction parameters and growth rates tailored precisely to that individual patient's *in vivo* physiology. These parameters learned in one data set can also be used as parameter range estimates for other data sets where longitudinal data is not available. Alternatively, when longitudinal clinical data is absent from the beginning, we will use alternative parameterization paths to either create *in silico* patient cohorts through Latin Hypercube Sampling (LHS), as done previously (39), or use empirical scaling of parameters in the LV framework based on published *in vitro* co-culture experiments that inform interaction terms (40). These alternative approaches ensure our model can perform broad predictive screening of drug efficacy and hormone fluctuations across diverse virtual phenotypes based on snapshot baseline compositions.

Open model infrastructure and sharing

To make **MOSAIC** more than a one-off integrated model, we will build it around open, modular software infrastructure for composing, sharing, and coordinating model development across teams. Each model component will have an explicit interface that defines its inputs, outputs, assumptions, units, timescale, and biological scope, allowing microbiome dynamics, metabolic models, agent-based models, and whole-body compartment models to be developed and validated independently while still working together in larger simulations. For general data sharing, we will use common tools for which all our institutions have licenses (e.g., Box). We will use GitHub-based infrastructure for version control, issue tracking, documentation, code review, and coordination across modeling teams, and distribute reusable software components through PyPI where appropriate. Models, parameter sets, simulation workflows, and validation datasets will be packaged with clear provenance and documentation so they can be reused, extended, and compared across teams. We will consider personal health information (PHI) and try to be as streamlined as possible, while also taking university policies into account. This open infrastructure will make **MOSAIC** more reproducible, easier to validate, and adaptable beyond the initial use cases.

Advantages & strengths

As already highlighted above, **MOSAIC** embraces the inter-system complexity and applies the appropriate modeling framework to the associated system rather than forcing the biology to fit into a specific modeling framework. There are apt analogies in many high-functioning engineered systems; an airplane uses optimal materials for its outer shell to maximize aerodynamics and hull integrity and different optimal materials for seat covers to maximize passenger comfort. Similarly, **MOSAIC** capitalizes on the best modeling framework for a given physiological system and then identifies mechanisms of interconnectivity. All done with state-of-the-art model construction and coding practices.

Accounting for sex as a biological variable

Extensive research demonstrates that sex-specific physiological differences significantly influence therapeutic efficacy and toxicity. Building on this, previous work from the Papin lab (41) utilized sex-specific transcriptomic data to reveal distinct metabolic variations, particularly within the pentose phosphate pathway, between male and female genome-scale liver metabolic models. These metabolic differences are hypothesized to modulate individual susceptibility to hepatotoxic pharmaceuticals.

MOSAIC proposes to bridge the gap in existing literature surrounding the mechanistic role of hormone homeostasis in shaping therapeutic response. **MOSAIC** will account for sex as a biological variable through hormone profiling and the incorporation of sex-specific physiology. **The novelty lies in the fact that rather**

than looking at hormone levels in isolation, MOSAIC will assess how hormone dynamics bidirectionally affect metabolism and host-microbiome interactions – all of which integrate to shape an individual's therapeutic response. While current computational frameworks model the gut, liver, and vaginal systems in isolation, our integrated approach captures how sex-specific variables, such as estrogen dynamics, drive disease phenotypes in BV and MASLD. Within this framework, PKPD models will track systemic hormone and drug kinetics to inform constraints for metabolic network models, enabling highly accurate and physiologically relevant metabolic flux predictions. These sex-specific metabolic profiles will define the functional parameters for community and agent-based models. Crucially, this framework establishes a bidirectional relationship where sex-specific differences actively drive the model calibration, whereby population and spatial dynamics continuously re-parameterize the metabolic network and PKPD models. In addition, the PK of drugs and other therapies can be sex-specific, and this will be included where relevant based on clinical pharmacometric studies that measured these differences (42,43).

Assessing the model's validity and credibility

Model credibility will be assessed at multiple levels, including biochemical and physiological consistency, mass balance and feasible flux distributions, and agreement with known sex-specific patterns of metabolism and hormone regulation. Parameter identifiability methods, both structural and practical, will be used to ensure that the data being used to parameterize the model is sufficient for the purpose. Several predictions of the modules will be directly compared to accessible and repeatable clinical measurements, e.g. hormone levels in the blood before and after interventions and plasma drug concentrations over time, and this data will be separated into training data for parameterization and testing data for validation, with no overlap between the two. Uncertainty quantification and uncertainty propagation methods will be used to quantify the robustness of the predictions of the linked models to variability in the input data. Relevance will be assessed by comparing model behavior with independent human datasets associated with MASLD and BV. AI-based sensitivity analyses will be used to identify key drivers, uncertainties, and limitations, informing iterative model refinement.

Scientific impact and clinical relevance

The successful implementation of the proposed work will result in a harmonized computational modeling platform using key modeling approaches. **MOSAIC** will enable identification of potential therapies for metabolic and microbial targets and prediction of adverse drug effects in a sex- and disease-specific manner. Because the approach is also human-focused, results are directly translatable for clinical considerations and therapeutic decisions. Specifically, **MOSAIC** will enable the prediction of the efficacy for therapeutics (drugs or biologics) for BV, which affects millions of women per year, with highly variable rates of responsiveness and recurrence and in which pre- and postmenopausal women have different manifestations. **MOSAIC** will also enable the prediction of therapeutic efficacy of emerging drugs for MASLD which has notable sex-specific differences. Both conditions have strong metabolic and microbiome factors. Both conditions are highly dependent on hormone homeostasis. Importantly, while BV and MASLD are the initial focus, because **MOSAIC** will be developed with best-practices in software design and model construction, the framework will be readily extensible to other diseases. We will focus on predictions of efficacy and adverse effects for the therapeutics metronidazole and *L. crispatus* for BV and resmetirom and GLP-1 RA for MASLD as starting points, with intention to generalize the approach to other interventions as feasible.

Experience and past success of the project team

All Investigators bring extensive experience leading large, complex research teams, with past leadership highlights and the coordination of **MOSAIC** personnel detailed in the Leadership Plan. Each lab will dedicate at least one trainee or staff member to each of the four teams (see Leadership Plan) to ensure cohesive integration and fluid data exchange across projects. Many **MOSAIC** investigators have served as PIs or Directors of large scientific activities (see Biosketches and Leadership Plan). There is a history of collaboration among all members of the **MOSAIC** team.

The team of investigators collectively provides expertise in genome-scale metabolic modeling (Papin), microbiome ecology (Arnold), PKPD modeling (Mac Gabhann), agent-based modeling (Peirce-Cottler), software infrastructure (Agmon), reproductive biology (Derré), liver physiology (Khetani), microbiome-focused clinical research (Young, Vaughan), and experimental validation systems (Papin, Peirce-Cottler, Derré,

Vaughan, Young, Khetani). Together, the team has expertise with each modeling modality represented in **MOSAIC**.

Potential risks and alternative plans

Risks for the proposed modeling efforts include #1) paucity of appropriate experimental data to properly parameterize the models, #2) model predictions that fail validation experiments, and #3) lack of connectivity between modeling frameworks. By design, the **MOSAIC** team is composed of cell biologists and clinicians with the relevant expertise. This expertise ensures that the models are grounded in biological and clinical reality. In addition, these investigators readily generate the data types that the models will use. For risk #1, if existing data is insufficient, we are well positioned to generate the data ourselves (see below). Validating experiments often identify aspects of the model that fail and in fact such failures are often a catalyst to the discovery of new biology. For risk #2, we will approach consistent model prediction failures by limiting the scope of the model (e.g., focus on liver metabolism alone instead of coupling with the role of microbiome metabolism) and pivot the therapeutic focus to drugs that have a more targeted effect. For risk #3, integrated modeling approaches can face challenges with differences in spatial and temporal scales. If we find that the dynamics and parameters that are critical at the interface of the modeling approaches are difficult to identify, we can leverage recently developed AI tools (like graph neural networks (44)) that can extract input/output relationships without explicit accounting for underlying mechanisms. While our preference is to keep many of the underlying models mechanistic because doing so allows us to identify specific molecular targets, we can consider applying graph neural networks or other such approaches to abstract some of those connections to ensure model functionality.

Reuse of existing resources

MOSAIC will serve as an excellent platform to integrate existing multiomics data (metabolomics, transcriptomics, genomics, proteomics) and hormonal and drug dynamic information. The **MOSAIC** investigators are tightly connected with many of the data resources described below as either collaborators with the associated PIs or as contributors to the actual data repositories.

Metagenomics and multiomics of microbial communities

The University of Maryland Baltimore, Human Microbiome Project (UMB-HMP) dataset (BioProject PRJNA208535) contains about 5000 metagenomic profiles of the vaginal microbiome in the absence and presence of menses. The original cohort data contains vaginal swabs from 135 nonpregnant women of reproductive age, collected daily for 10 weeks (45). The dataset has since been expanded to include shotgun sequencing of several thousand vaginal samples. This large dataset will be used to create personalized digital twins of individual vaginal microbiomes and train machine learning models to predict vaginal microbiome compositions over time.

The multi-strain synbiotic vaginal tablet formulation (VS-01) dataset includes shotgun metagenomic profiles of vaginal microbiomes from participants receiving either a probiotic intervention or placebo (46). The CONRAD BV dataset includes 16S metagenomic profiles from participants with bacterial vaginosis (BV) who received metronidazole treatment (47) while the VIBRANT dataset includes shotgun metagenomic profiles from participants with BV who received metronidazole treatment followed by either a probiotic intervention or placebo (48). Metatranscriptomics of the vaginal microbiome in BV patients treated with metronidazole is available under PRJEB21446 and will be used to further profile models of the vaginal microbiome (49). Metabolomics of the vaginal microbiome in women with and without BV will be used to parameterize metabolic conditions in the vaginal environment (50). Together, these datasets provide longitudinal profiles of vaginal microbiome responses to therapeutic interventions and will be used to train and evaluate machine learning models for predicting microbiome dynamics following treatment.

The Human Microbiome Project (HMP) dataset will also be used to generate *in silico* clinical cohorts representative of the healthy gut microbiome. The HMP1-II project (PRJNA48479) contains 249 metagenomic profiles from healthy individuals and provides a cross-sectional characterization of gut microbial composition (51). The Broad Institute–OpenBiome Microbiome Library (BIO-ML) (PRJNA544527) includes longitudinal stool metagenomic data from 80 healthy participants (52). Metabolomics were also run on samples collected at various time points over the 18 months. The BIO-ML dataset will complement the HMP cohort by providing longitudinal data to capture temporal variability in the healthy gut microbiome.

A multiomics dataset, including 16S metagenomics, transcriptomics, and metabolomics of stool samples from NAFLD-diagnosed and control participants will be used to model the gut microbiome in

individuals with and without MASLD (53). The dataset includes samples from men and postmenopausal women. A larger dataset (PRJNA629344) containing 16S and shotgun metagenomics of MASLD and control participants can be used to increase the power of metagenomic analysis if needed (54).

Host cell transcriptomics

The BARICO dataset (PRJNA1051582) provides bulk RNA-seq transcriptomic profiles from liver biopsies of individuals diagnosed with metabolic dysfunction-associated steatotic liver disease (MASLD) (17). An additional study (PRJNA1154708) includes liver transcriptomic profiles and plasma proteomic profiles from participants with MASLD and healthy controls (55). Both datasets include male and female representation. Together, these datasets provide complementary host molecular profiles that will be used to investigate host alterations associated with MASLD. RNAseq transcriptomic data (PRJNA731542) from ectocervical biopsies of individuals with BV that have been given metronidazole will be used to profile cervicovaginal cell models (56). A subset of the HuBMAP collection (HBM692.JRZB.356) will be used to profile gut epithelial cell models of healthy donors. This dataset includes scRNA-seq transcriptomics of 8 different collection sites in the gut (57). There are currently no published datasets of gut epithelial transcriptomics in individuals with BV or MASLD.

Hormone profiling and other data sets

The mcPHASES dataset will be used to model levels of estrogen and progesterone during the menstrual cycle (58). mcPHASES includes daily urinalysis of hormone metabolites across a three-month observation period. All data is publicly available on Physionet. The MESA AC 291 dataset (BioLINCC HLB00640825a) will be used to analyze levels of bioavailable testosterone, estradiol, dehydroepiandrosterone and sex hormone binding globulin associated with NAFLD (59). This subset of MESA includes data from postmenopausal women and men. We will use published pharmacokinetic (PK) profiles of resmetirom (42) and metronidazole (43) to inform model parameterization and drug exposure dynamics. We will use AERSMine (60), an application which makes the Food and Drug Administration's Adverse Event Reporting System (AERS) more accessible, as we have done in the past (41). AERSMine can be used to analyze trends in side effect reports by specific drug, adverse event, sex, and age.

Primary data collection

All primary data collection will be model-driven, restricted to parameterization gaps or independent validation of model predictions. We plan to use the human-centered *in vitro* models outlined below to generate these data where the publicly available datasets do not provide all of the data necessary for model input or where specific validations will be performed (e.g., drug testing, hormone levels). Experiments with such i*n vitro* systems can be integral to identifying areas in the models where further curation may be required to improve predictions. Regarding human subject research, an observational retrospective study to collect vaginal swabs is already IRB approved and active at UVA. Through this study, we can collect clinical data through electronic medical records and match vaginal swab samples. The current study is limited by being a cross-sectional study and there is no patient contact. If there is a need to develop a prospective longitudinal study to gather samples that can better inform the models (over *in vitro* approaches), then we will develop and go through the approval process. A prospective longitudinal study has the potential to provide valuable data as to the personalization of the **MOSAIC** platform.

Rigorous model validation will require independent experimental testing. A key characteristic of the **MOSAIC** modeling framework is to embed a person with each type of modeling expertise (metabolic network models, community models, agent-based models, PKPD models, multi-scale models) on each application team (BV, MASLD) to ensure there is actual crosstalk between the modeling frameworks. In addition, each application team (BV, MASLD) has a cell biologist and a clinician with the respective expertise to ensure models are constructed with awareness of the underlying biology and relevant drug targets from the very beginning of model construction.

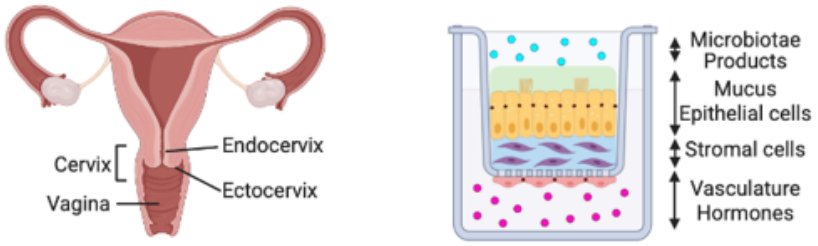


Figure 3. An *in vitro* model of the female reproductive tract consists of epithelial and stromal cells interfacing microbial and hormonal components. Derré has developed this cell culture system with remarkable consistency with human physiology. This system can serve as an important testbed for model parameterization and validation as necessary.

Reproductive Tract Model (Figure 3): To model the effects of hormonal regulation within the human female reproductive tract, we established a stepwise multicellular Transwell platform composed of human vaginal or endocervical epithelial cells (cell lines or primary cells) cultured on the apical surface of a collagen matrix containing SHT290 fibroblasts. The system can be further expanded by incorporating endothelial cells on the basolateral side of the membrane to increase physiological complexity. Following establishment of the model, hormones can be administered to the basolateral compartment to mimic systemic exposure, while microbiota-derived products or live bacteria can be applied to the apical epithelial surface to model host–microbe interactions. This modular platform allows the addition or removal of specific cell types as needed and is readily compatible with a range of downstream analyses (61,62). Potential therapeutics to be examined in the *in vitro* reproductive tract model include the standard of care for BV (metronidazole) and a biologic/probiotic (*Lactobacillus crispatus*) which have been used in clinical trials (63,64).

Human 3D liver microtissues (Figure 4)*:* The Khetani lab utilizes droplet microfluidics to fabricate extracellular matrix protein-based microtissues containing primary human hepatocytes (PHHs) (65–67) with hepatic stellate cells (HSCs), or liver sinusoidal endothelial cells (LSECs). Microtissues conserve the use of limited primary human liver cells, and exhibit 3D cell-cell and cell-ECM interactions, unlike 2D monolayers used in existing liver-on-a-chip (microfluidic) devices (68–73). Using various combinations, noninflamed and inflamed (MASLD) states of the microtissues can be generated as shown by high levels of inflammatory cytokines and other inflammatory markers that are consistent with clinical studies in MASH patients due to inflammation (74,75).

Gut-liver model (Figure 4)*:* Similar to the reproductive tract model, a transwell configuration using primary human intestinal stem cells (membrane layer to establish polarity) and fibroblasts and PHHs (bottom of well) can be used to examine how the intestinal microbiome interacts with the liver and gut cells using TEER (transepithelial resistance) and CYP3A4 levels to monitor intestine and liver cell activity (76–80). Two FDA approved drugs of interest to test *in vitro* include resmetirom and GLP-1 receptor agonists. Resmetirom is a thyroid hormone receptor beta agonist and is used in adults with noncirrhotic MASH (81,82). Use of the Glucagon-like peptide-1 receptor agonist (GLP-1 RA, semaglutide) in patients with MASLD and MASH exhibited sex-dependent and independent differences in responses (8,83). The hepatic benefit is largely mediated through improved insulin sensitivity, reduced delivery of lipotoxic substrates, and the gut-liver-microbiome axis (84). Overall, MASLD is sexually dimorphic, with the estrogen, androgen, and thyroid axes each modulating hepatic lipogenesis, fatty acid oxidation, and fibrogenesis likely affecting gut epithelial cells and the microbiota.

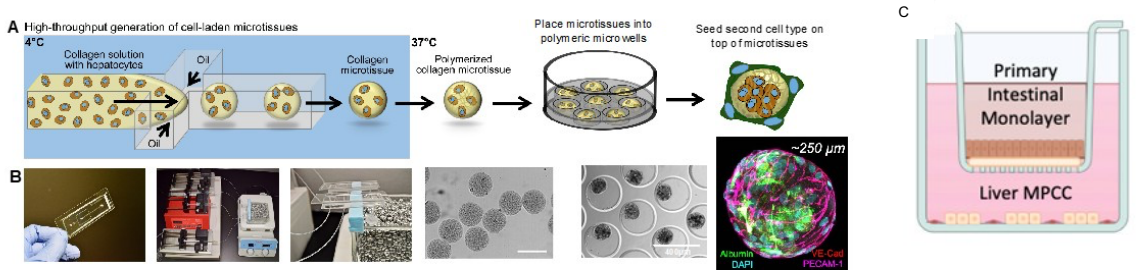


Figure 4. (A) To represent liver tissue function, a mixture of ECM and cells is perfused into a microfluidic droplet generator at specific experimental conditions to enable microstructures consistent with liver tissue physiology (**A**). Polydimethylsiloxane (PDMS) microfluidic devices with a single emulsion droplet generator to facilitate the creation of the liver tissue models **(B).** An intestinal epithelial monolayer with cocultured micropatterned cells (MPCC) in a Transwell format to simulate the gut influence on liver function (**C**).

Conclusion

MOSAIC is a potential paradigm-shifting strategy for modeling human physiology. Rather than developing a single modeling framework to represent varied biology, **MOSAIC** embraces the reality of interconnected systems that underlie health and disease. Hormone homeostasis is embedded in every facet of **MOSAIC.** For example, transcriptional profiles as a function of estrogen levels will be integrated into the metabolic network models, PKPD models will explicitly account for estrogen dynamics, microbial community models will be derived from data that captures dynamics of menses, and models will be constructed as a function of hormonal status (e.g., pre/post menopause). The team of investigators possess the ideal combination of modeling expertise and associated biological and clinical expertise. **MOSAIC** will be well-positioned to predict sex-specific therapies for BV and MASLD and lay the foundation for application to other diseases at the interface of human metabolism and the microbiome.

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Resource Sharing Plan

The MOSAIC team is committed to rapid dissemination of models, software, datasets, and documentation to maximize transparency, reproducibility, and community reuse. Importantly, during the model integration phase, we anticipate having several people across the modeling efforts that can cross-check/validate code function. A MOSAIC GitHub will be used for active development, version control, issue tracking, collaborative development, and dissemination (see "*Computing Infrastructure*" in the Equipment and Facilities Section and "*Public repositories, Software and Licensing, and Preservation and Timelines"* in the Data Management/Sharing Section).

Model Components and Modeling Parameters: The computational models/modeling components will be made available through community repositories as available (e.g., Biomodels at EBI). The standard format for such computational models is SBML (systems biology markup language). Models will also be made freely available through public GitHub repositories, as well as through supplementary material in accompanying publications. Documentation detailing model specification, parameter sets, simulation configuration/results, metadata, and other datasets will be freely available through project repositories and associated documentation sites when appropriate. In addition, software and algorithms that are developed during this project will be made available with appropriate repositories (e.g., GitHub). We will similarly make newly developed computer code available. Resources not suitable for repository deposition will be distributed upon reasonable request, consistent with institutional, ethical, and regulatory requirements. The investigators have a strong track record of publicly disseminating computational models and software. For example, Papin and Mac Gabhann co-authored the code availability policy of PLOS Computational Biology {PMID 35653366} which has been championed across the publishing industry.

Associated Scientific Data: The use of publicly available data will be cited with the corresponding models and include all relevant metadata. Any nucleic acid sequence read data generated uniquely through this project will be annotated and deposited to NCBI *Sequence Read Archive*. High-throughput metabolomic data will be uploaded to *MetaboLights*, an open-access metabolomic data repository by European Bioinformatics Institute.

In the case where samples may be shared as dictated by approved IRB protocols, material availability, and in accordance with the proposed work, we would follow all shipping regulations and use the Uniform Biological Material Transfer Agreement and/or consult with University counsel regarding any materials transferred.

Documentation and reproducibility: The MOSAIC team members have championed the reproducibility of computational models {PMID 32845080, 32427998, 41343683}. We will implement practices to ensure that there is sufficient documentation and metadata to enable model reproduction and independent validation. All code development will occur with field standards in version control and collaborative coding efforts. By embedding personnel from each modeling-focused lab across the BV, MASLD, Integration, and Software teams, we will promote cross-laboratory awareness, compatibility, and early identification of reproducibility issues. All software and computer models will be checked by the Software team (Agmon) for criteria of proper documentation and delineation of software dependencies. Agmon was a member of the Center for Reproducible Biomedical Modeling {PMID 41343683} and he will ensure that MOSAIC is implementing best practices.

Timelines: Curated datasets, models, software releases, and documentation supporting publications will be publicly released at the time of publication, with interim releases made whenever feasible to accelerate community use and feedback. GitHub will be used for active development, while stable releases and associated datasets may be archived through Zenodo or similar long-term repositories with persistent identifiers. We will not delay disclosure of research findings for the purpose of evaluating an invention for a patent application or the actual filing of a patent application.

Open model infrastructure and sharing: To make MOSAIC more than a one-off integrated model, we will build it around open, modular software infrastructure for composing, sharing, and coordinating model development across teams. Each model component will have an explicit interface that defines its inputs, outputs, assumptions, units, timescale, and biological scope, allowing microbiome dynamics, metabolic models, agent-based models, and whole-body compartment models to be developed and validated independently while still working together in larger simulations. Consistent with the goals of OTA-26-004, MOSAIC models, software, and supporting resources will be designed to enable reuse, extension, and integration by external investigators studying related hormone-regulated biological systems. Internal project collaboration will use institutionally approved file-sharing platforms. Public dissemination will occur through established repositories appropriate to the resource type. We will use GitHub-based infrastructure for version control, issue tracking, documentation, code review, and coordination across modeling teams, and distribute reusable software components through PyPI where appropriate. Models, parameter sets, simulation workflows, and validation datasets will be packaged with clear provenance and documentation so they can be reused, extended, and compared across teams. Sharing of human-subjects data will comply with IRB approvals, informed consent provisions, HIPAA requirements, and institutional policies governing protected health information. This open infrastructure will make MOSAIC more reproducible, easier to validate, and adaptable beyond the initial use cases.

Data Management and Sharing Plan

Data Types: The MOSAIC project will generate scientific data, software, documentation, and training materials related to various types of computational modeling and simulation across the investigators. Expected computation-related outputs include model specifications, simulation configurations, parameter sets, structured simulation results, metadata, schemas, analysis scripts, notebooks, APIs, source code, documentation, tutorials, and example datasets. Expected biological data related outputs include genomics, 16S sequencing, transcriptomics, metabolomics, and other cell based measures.

Reproducibility and Validation: Sharing these outputs will support transparency, reproducibility, independent validation, and reuse by the broader modeling and scientific communities. Metadata and documentation will describe data formats, simulation parameters, model assumptions, workflows, and methods needed to reproduce published results. Any specialized tools, software, or code needed to access or manipulate the shared scientific data will be readily provided through the computational model repositories or supplemental files of manuscripts with links to community tools that are freely accessible. Validation datasets, benchmark results, and associated metadata used to assess model performance will be shared whenever possible to facilitate independent evaluation of model credibility.

Model and Simulation Standards: Model-related outputs will be shared using community standards whenever possible. Models will be represented in SBML where appropriate, including relevant parameters, environmental conditions, and simulation assumptions. Simulation metadata, experimental designs, and workflow specifications will be described using SED2, a community standard for describing simulation experiments and workflows, when applicable.

Public Repositories: Curated models suitable for broader reuse will be deposited in public repositories such as BioModels, and published simulation results may be shared through BioSimulations when appropriate. GitHub will be used for active development, version control, issue tracking, collaborative development, and dissemination. Any nucleic acid sequence read data generated uniquely through this project will be annotated and deposited to NCBI *Sequence Read Archive*. High-throughput metabolomic data will be uploaded to *MetaboLights*, an open-access metabolomic data repository by European Bioinformatics Institute. The metadata will also be linked as appropriate by filename to the figures of the associated preprints/publications/repository files.

Software and Licensing: Custom software, schemas, workflows, and analysis tools will be released through public GitHub repositories under permissive open-source licenses such as Apache 2.0, MIT, or BSD, unless project-specific constraints require otherwise. Repositories will include README files, license information, citation guidance, installation instructions, examples, tests where feasible, and documentation for both users and developers.

Formats and FAIR Principles: All labs will prioritize open, machine-readable, and interoperable formats, including SBML, SED2, JSON, YAML, CSV, HDF5, and other community-supported formats as appropriate. The MOSAIC team will follow open-science and FAIR principles to maximize findability, accessibility, interoperability, and reuse.

Preservation and Timelines: Curated datasets, models, software releases, and documentation supporting publications will be made publicly available as soon as possible. GitHub will be used for active development, while stable releases and associated datasets may be archived through Zenodo or similar long-term repositories with persistent identifiers. All scientific data generated (experimental, observational, derived from public repositories, simulations) will be made publicly available before a formal patent application is filed. Data sharing will not be delayed for the purpose of evaluating an invention for patent application or filing a patent application.